

Notes: Kerr & Nanda (JFE, 2009)

Democratizing Entry: Banking Deregulations, Financing Constraints, and Entrepreneurship

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	3
3.1	Banking deregulations	3
3.2	Longitudinal Business Database and entrant types	4
3.3	Entry, exit, and concentration outcomes	4
4	Models and empirical specifications	4
4.1	State-year entry and exit panel	4
4.2	Relative start-up design	4
4.3	Industry external-finance interaction	5
4.4	Size and concentration tests	5
5	Identification and empirical design	5
5.1	Core identification logic	5
5.2	Why interstate reforms matter most	5
5.3	Addressing confounds	5
5.4	Limitations	6
6	Tables and figures	6
6.1	Fig. 1: Cumulative states passing US branch banking deregulation; Table 1: Descriptive statistics on changes in the US banking industry	6
6.2	Table 2: Descriptive statistics on new establishments; Fig. 2: US establishment entry patterns by entrant type	6
6.3	Fig. 3: Intrastate and interstate reform timing	7
6.4	Table 3: Entry and exit around deregulation; Fig. 4: Dynamic intrastate and interstate effects	7
6.5	Table 4: Dynamic panel estimations of establishment entry and exit	7
6.6	Table 5: Relative entry count differences-in-differences; Table 6: Dynamic relative entry design	9
6.7	Tables 7 and 8: Size distribution of churning and long-term entry	9
6.8	Tables 9 and 10: Incumbent market shares and concentration	10
6.9	Fig. A1: Single-unit and multi-unit dynamics	11
7	Short summary	11
8	Conclusion	12
8.1	Core conclusion	12
8.2	Economic intuition	12

8.3	Incremental contribution revisited	12
8.4	Implications	12
8.5	Limitations	12
8.6	Takeaway	12

1 Big picture

- **Question.** How did U.S. banking deregulations affect entrepreneurship, firm entry, closures, and creative destruction?
- **Policy shocks.** The paper studies intrastate and interstate branch banking deregulations across U.S. states. Intrastate reforms eased branching within a state; interstate reforms allowed cross-state banking expansion.
- **Data.** The main data come from the Census Longitudinal Business Database (LBD), which tracks establishment births, deaths, ownership status, employment, and survival.
- **Main result.** Interstate deregulation produced large increases in new business entry, but much of the increase came from *churning entrants*: start-ups that closed within three years.
- **Creative destruction nuance.** The deregulations did increase entry and some displacement of incumbents, but the largest effect was democratized experimentation: many new firms entered, many failed, and a smaller set survived.
- **Why this matters.** Better capital markets do not simply select the best entrepreneurs ex ante. They may also allow many more entrepreneurs to try, generating failure as an essential part of the process.
- **Economic takeaway.** Financial development democratizes entrepreneurship by relaxing entry barriers, but creative destruction requires both successful new firms and a large amount of failed experimentation.

2 Incremental contribution of the paper

- **From finance-growth to firm demographics.** Prior work links financial development to economic growth. This paper opens the black box by studying establishment entry, survival, and exit directly.
- **Separates entry quality from entry quantity.** The paper distinguishes all entry, churning entry, and long-term entry. This separation reveals that financial reform produces many short-lived firms, not just a smooth increase in high-quality entry.
- **Uses establishment-level Census data.** Instead of relying only on aggregate state data or Compustat firms, the paper uses the LBD to capture small start-ups, business closures, and survival durations.
- **Links banking deregulation to product-market dynamics.** The paper connects bank-market reforms to Schumpeterian creative destruction, but shows that the empirical process is messier than a simple replacement of bad incumbents by good entrants.
- **Clarifies why earlier findings differ.** The churning result helps explain why previous studies find strong entry effects after deregulation but weaker effects on firm-size distributions or aggregate growth.
- **Democratization interpretation.** The paper reframes financial development as increasing access to entrepreneurial experimentation, not merely reallocating capital perfectly to ex ante best projects.

3 Data and measurement

3.1 Banking deregulations

Two reforms are central. Intrastate deregulation lets banks expand branch networks within a state. Interstate deregulation permits banking organizations to expand across state borders. Fig. 1 shows that interstate reforms were concentrated during the 1980s and early 1990s, overlapping the LBD sample window.

Interpretation: The staggered timing across states creates the policy variation. The paper treats reforms as shocks to local bank competition and the supply of entrepreneurial finance.

3.2 Longitudinal Business Database and entrant types

The LBD tracks U.S. establishments over time. The analysis excludes the financial sector and focuses on establishment entry and closure outside banking. The main entrant categories are:

- **New single-unit start-ups:** new independent firms.
- **Multi-unit facility expansions:** new establishments opened by existing firms.
- **Churning entrants:** entrants that close within three years.
- **Long-term entrants:** entrants that survive four or more years.

Interpretation: The contrast between new single-unit start-ups and multi-unit facility expansions is crucial. Facility expansions serve as a comparison group for establishment formation by existing firms, while start-ups are the entrepreneurship outcome.

3.3 Entry, exit, and concentration outcomes

The paper studies entry counts, churning entry, long-term entry, establishment closures, size distribution at entry, and incumbent concentration. Incumbent concentration is measured by the top-ten market share and normalized Herfindahl-Hirschman index at the state-industry-year level.

Interpretation: These outcomes separate two mechanisms. Entry and survival measure entrepreneurship; closures and concentration measure creative destruction and competitive displacement.

4 Models and empirical specifications

4.1 State-year entry and exit panel

The baseline state-year specification estimates effects of reforms on log entry and closure counts:

$$\log(BIR_{s,t}) = \alpha_s + \gamma_t + \beta_I Instate_{s,t} + \beta_E Interstate_{s,t} + X'_{s,t} \delta + \varepsilon_{s,t}.$$

Dynamic versions use leads and lags around reforms, with the omitted period being two years before reform or earlier.

Interpretation: A positive interstate coefficient on start-up entry means deregulation increased new firm formation. Positive effects on churning entry indicate more failed experimentation, not necessarily higher-quality selection.

4.2 Relative start-up design

To isolate start-up effects from general state-year shocks, the paper compares start-ups with facility expansions inside state-industry-type-year cells:

$$\ln(BIR_{s,i,t,x}) = \phi_{s,i,x} + \tau_{t,x} + \eta_{s,i,t} + \beta_I Instate_{s,t} \times Type_x + \beta_E Interstate_{s,t} \times Type_x + \varepsilon_{s,i,t,x}.$$

Here $Type_x$ indicates whether the entrant type is a start-up rather than a facility expansion.

Interpretation: This is a difference-in-differences within local industry cells. It asks whether start-ups grow more than expansions by existing multi-unit firms after deregulation.

4.3 Industry external-finance interaction

Some specifications interact deregulation with industry dependence on external finance:

$$\ln(BIR) = \dots + \theta Reform_{s,t} \times ExternalFinance_i + \varepsilon.$$

Interpretation: If deregulation relaxes financing constraints, effects should be stronger in industries more dependent on outside finance. The paper uses this as mechanism evidence.

4.4 Size and concentration tests

The paper estimates entry effects by entrant employment size and studies incumbent market share/concentration. These tests separate an extensive margin of more firms from an intensive margin of larger entrants and a creative-destruction margin of incumbent displacement.

5 Identification and empirical design

5.1 Core identification logic

The design relies on staggered state-level bank deregulation. States deregulate at different dates, and establishment entry/exit is observed before and after reform. State fixed effects absorb permanent state differences; year fixed effects absorb national shocks.

Interpretation: The baseline compares treated states to not-yet-treated states over time. Dynamic event-study specifications check for pre-trends and show how effects evolve after reform.

5.2 Why interstate reforms matter most

The paper finds stronger and more persistent effects for interstate deregulation than intrastate deregulation. Interstate reforms intensified competition and integration across state banking markets, making credit more broadly available and pushing banks toward larger, more efficient structures.

5.3 Addressing confounds

- **Pre-trends.** Fig. 3 shows little relation between pre-existing entry rates and deregulation timing.
- **State-year shocks.** Relative start-up models include state-industry-year fixed effects.
- **Industry mechanisms.** External-finance interactions test whether more finance-dependent industries respond more.
- **Survival.** Churning versus long-term entry separates experimentation from durable entry.
- **Incumbent effects.** Closure and concentration tests evaluate creative destruction rather than only start-up formation.

Interpretation: The paper's strongest design is the relative start-up specification because it compares start-ups against facility expansions in the same state-industry-year environment.

5.4 Limitations

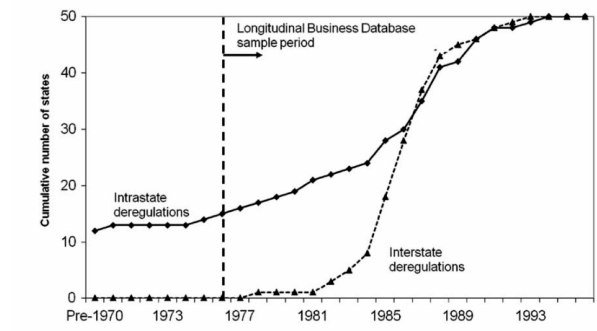
The paper cannot observe loan-level finance for each start-up. Deregulation could also affect product demand, local competition, or bank consolidation in ways not fully captured by fixed effects. The results are therefore interpreted as reduced-form effects of banking deregulation on product-market entry and exit.

6 Tables and figures

6.1 Fig. 1: Cumulative states passing US branch banking deregulation; Table 1: Descriptive statistics on changes in the US banking industry

Read: Fig. 1 shows the timing of intrastate and interstate deregulation. Table 1 shows that banking became more consolidated, with fewer banking organizations and more branches, especially out-of-state branches.

Economic message: The banking reform environment changed both competition and geographic reach. Entry effects are interpreted against this banking-sector restructuring.



ates passing US branch banking deregulation. This figure plots the cumulative number of states passing the

Fig. 1: cumulative states passing branch banking deregulation.

Deregulation timing and banking-sector restructuring. Interstate liberalization rises sharply during the LBD sample period, while bank consolidation and interstate branching increase.

Descriptive statistics on changes in US banking industry from 1977 to 1994. Panel A shows data from the Longitudinal Business Database (LBD) for banking industry (602 in 1987 Standard Industrial Classification). Panel B shows asset-based descriptive statistics taken from Berger, Klapper, and Udell (2001).

Panel A. Change in number of banks and bank branches		
	1977	1994
Total number of banking organizations	12,810	8,547
Percent with fewer than 50 employees (%)	79	70
Percent with branches in multiple states (%)	0	3
Total number of banking branches	38,231	64,155
Percent owned by banks with 500+ employees (%)	50	66
Percent owned by banks with mean 500+ employees before 1985 (%)	52	62
Percent owned by banks originally located in other states (%)	2	25
Panel B. Change in bank assets		
	1979	1994
Total number of banking organizations	12,463	7,926
Percent small banks (less than \$100 million in assets) (%)	80	71
Real gross industry assets (in trillions of 1994 dollars)	3.26	4.02
Percent assets in megabanks (more than \$100 billion in assets) (%)	9	19
Percent assets in small banks (less than \$100 million in assets) (%)	14	7

Table 1: changes in banks, branches, and assets.

6.2 Table 2: Descriptive statistics on new establishments; Fig. 2: US establishment entry patterns by entrant type

Read: Table 2 shows that 82 percent of entering establishments are new single-unit start-ups; 42 percent of entrants churn within three years. Fig. 2 shows start-ups declining early and then partially recovering, while multi-unit facility expansions grow more rapidly.

Economic message: The raw data already show the importance of failure and entrant type. Start-ups are the bulk of entry, and a large share of them close quickly.

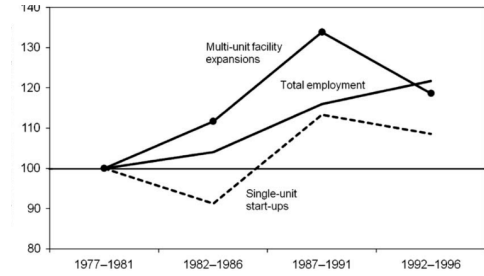
Table 2

Table presents descriptive statistics on the formation of new establishments outside of the financial sector using data from the Longitudinal Business Database (LBD). Statistics are calculated for entrants between 1977 and 1998 using data extending to 2001. Single-unit start-ups are new firm formations. Multi-unit facility expansions are new establishment openings by existing firms. Churning entrants are establishments closing within three years of entry. Long-term entrants are establishments surviving four or more years. Entry size distributions are calculated from year of establishment entry. Jarmin and Miranda (2002) describe the construction of the LBD. Sectors not included in the LBD are agriculture, forestry and fishing, public administration, and private households. We also exclude the US Postal Service, restaurants and food stores, hospitals, education services, and social services. These exclusions lower the services share relative to other sectors. Incomplete LBD records require dropping 25 state-year files: 1978 (12 states), 1983 (four), 1984 (four), 1985 (one), 1986 (one), 1989 (one), and 1993 (two).

	Entering establishments			Churning entrants (survive ≤ 3 years)			Long-term entrants (survive ≥ 4 years)		
	All entrants	New single-unit start-ups	Multi-unit facility expansions	All entrants	New single-unit start-ups	Multi-unit facility expansions	All entrants	New single-unit start-ups	Multi-unit facility expansions
Mean annual entry (thousands)	409	336	73	173	153	20	236	183	53
Share of entrants (%)		82	18	42	37	5	58	45	13
Size distribution									
1–5 employees (%)	70.3	76.0	44.1	76.3	79.5	51.6	65.9	73.0	41.3
6–20 employees (%)	22.8	19.7	36.9	18.6	16.9	31.7	25.9	22.1	38.9
21–100 employees (%)	5.8	3.8	14.9	4.4	3.2	13.3	6.9	4.3	15.6
100+ employees (%)	1.1	0.4	4.1	0.8	0.4	3.4	1.3	0.5	4.3
Sector distribution									
Manufacturing (%)	9	9	6	8	9	7	9	10	6
Services (%)	28	29	23	30	30	24	27	29	22
Wholesale trade (%)	12	11	17	11	10	18	12	12	17
Retail trade (%)	25	22	42	24	23	38	26	21	44
Mining (%)	1	1	1	1	1	1	1	1	1
Construction (%)	17	20	1	18	20	2	16	21	1
Transportation (%)	7	7	10	7	7	11	7	7	9
Geographic distribution									
Northeast (%)	19	20	17	18	18	17	20	21	17
South (%)	36	35	37	37	37	37	35	34	37
Midwest (%)	22	21	24	20	20	23	23	22	24
West Coast (%)	24	24	22	24	25	23	23	23	22

Table 2: entrant type, size, sector, and region distributions.

Descriptive entry patterns. The table highlights the prevalence of start-ups and churning; the figure compares start-ups, facility expansions, and total employment.



Entry patterns by entrant type. This figure plots establishment entry rates over the 1977–1996 period.

Fig. 2: entry trends by entrant type.

6.3 Fig. 3: Intrastate and interstate reform timing

Read: Fig. 3 plots pre-existing entry rates against deregulation timing. The flat fitted lines indicate little systematic relationship between pre-reform entry rates and reform dates.

Economic message: The figure supports the identifying assumption that deregulation timing was not strongly driven by pre-existing entrepreneurial activity.

6.4 Table 3: Entry and exit around deregulation; Fig. 4: Dynamic intrastate and interstate effects

Read: Table 3 shows that interstate deregulation increases start-up entry, churning entry, and long-term entry, while intrastate effects are weaker. Fig. 4 shows dynamic increases after interstate reform, especially for start-up entry and churning.

Economic message: The main result is not merely more entry; it is more entry and more short-lived failure. Banking reform democratizes attempts at entrepreneurship.

6.5 Table 4: Dynamic panel estimations of establishment entry and exit

Read: Table 4 reports dynamic leads/lags at the state-year level. Interstate reform effects are small before reform and rise in the years after reform, especially for new single-unit start-ups and churning entrants.

Economic message: The timing strengthens the causal interpretation: effects grow after deregulation rather than appearing before reform.

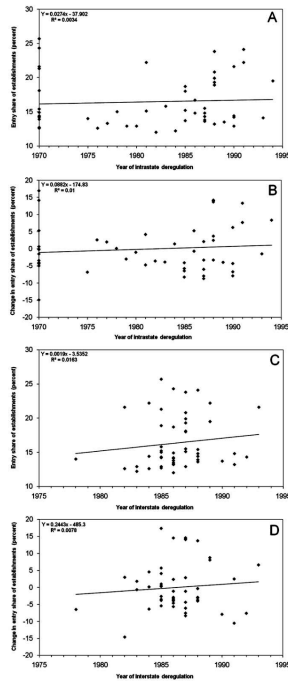


Fig. 3: Reform timing and pre-existing entry rates.

Reform timing, state-year entry/exit regressions, and dynamic effects on entry and closures. The evidence suggests no strong relation between pre-existing entry rates and deregulation timing, reducing concern about pre-trend selection.

Table 3
Panel estimates of establishment entry and exit at the state-year level using data from the Longitudinal Business Database (LBD) for 1977–1998. Coefficient reports estimates of the effect of interstate and interstate banking deregulations on the entry and exit of establishments. Dependent variables are indicated in column headers. Single-unit start-ups are new firm incorporations. Multi-unit facility expansions are new establishment openings by existing firms. Churning entrants are establishments closing within three years of entry. Long-term entrants are establishments surviving four or more years. The sample excludes all states and Washington, DC, merging 20 state-year cells in which LBD files are not available. For 1997 observations per regression, regressions include state and year fixed effects. Regressions include temporal interaction of explanatory indicators with a common year indicator. Regressions are weighted by average firm employment in states from 1977 to 1998. Standard errors are clustered at the state-year-occupation level. The post specifications compare annual entry rates before and after the state-level banking deregulation indicator. Linear treatment effect specifications allow for heterogeneous treatment effects over time by modeling the number of years after the indicated deregulation's passage, with a long-term effect at four years. The Appendix reports these estimations with linear state time trends incorporated.

	Log total entrants		Log churning entrants		Log long-term entrants		Log establishment closures	
	New single-unit start-ups (1)	Multi-unit facility expansions (2)	New single-unit start-ups (3)	Multi-unit facility expansions (4)	New single-unit start-ups (5)	Multi-unit facility expansions (6)	Single-unit firms (7)	Multi-unit facility closures (8)
Interstate banking deregulation post indicator	0.002 (0.004)	-0.010 (0.015)	-0.027 (0.021)	-0.087 (0.042)	0.027 (0.030)	-0.001 (0.033)	-0.009 (0.033)	-0.013 (0.033)
Interstate banking deregulation post indicator	0.060 (0.020)	0.012 (0.017)	0.051 (0.030)	0.044 (0.030)	0.071 (0.018)	0.028 (0.018)	0.018 (0.025)	-0.017 (0.019)
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,097	1,097	1,097	1,097	1,097	1,097	1,097	1,097
Panel B: Linear treatment effect specifications								
Number of years since interstate banking reform	-0.005 (0.005)	0.004 (0.011)	-0.007 (0.012)	0.004 (0.012)	-0.001 (0.009)	0.004 (0.009)	-0.004 (0.011)	-0.001 (0.009)
Number of years since interstate banking reform	0.000 (0.000)	0.026 (0.009)	0.063 (0.011)	0.037 (0.010)	0.040 (0.011)	0.022 (0.008)	0.037 (0.009)	0.006 (0.009)
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,097	1,097	1,097	1,097	1,097	1,097	1,097	1,097

Table 3: State-year entry/exit regressions.

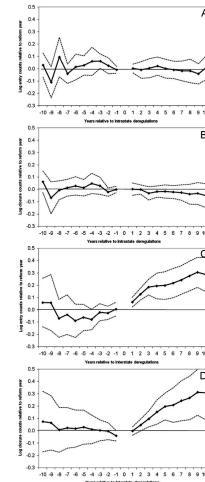


Fig. 4: Interstate and interstate dynamics for entry and closures. The figure plots coefficients from regressions of log entry counts or closure counts on a series of indicator variables extending from 10 years before the reform's passage to 10 years afterward. The end points include all earlier and later years. For each time variable, the first year of the indicator is defined as the coefficient on the corresponding variable in the year of the reform.

Fig. 4: Dynamic coefficients for entry and closures.

Table 4
Dynamic panel estimations of establishment entry and exit at the state-year level using data from the Longitudinal Business Database (LBD) for 1977–1998. Coefficients report estimates of the effect of interstate and interstate banking deregulations on the entry and exit of establishments. Dynamic specifications model a series of leads and lags for each reform. Leads and lags are consolidated into two-year increments extending from two years prior to the deregulations to four or more years after the deregulations. Coefficient values for dynamic leads and lags are relative to the period three years before reforms and earlier. The Appendix reports these estimations with linear state time trends incorporated. Table 3 provides further details on the specification.

	Log total entrants		Log churning entrants		Log long-term entrants		Log establishment closures	
	New single-unit start-ups (1)	Multi-unit facility expansions (2)	New single-unit start-ups (3)	Multi-unit facility expansions (4)	New single-unit start-ups (5)	Multi-unit facility expansions (6)	Single-unit firms (7)	Multi-unit facility closures (8)
Response to intrastate branch banking deregulations								
Years 1–2 before reform	-0.038 (0.034)	-0.015 (0.023)	-0.053 (0.041)	-0.023 (0.033)	-0.023 (0.030)	-0.009 (0.022)	-0.037 (0.026)	-0.001 (0.035)
Reform year and one after	-0.040 (0.044)	-0.016 (0.044)	-0.058 (0.057)	-0.012 (0.055)	-0.026 (0.036)	-0.016 (0.040)	-0.037 (0.042)	0.017 (0.043)
Years 2–3 after reform	-0.044 (0.035)	0.015 (0.042)	-0.063 (0.042)	0.027 (0.048)	-0.028 (0.036)	0.008 (0.044)	-0.050 (0.038)	0.020 (0.046)
Years 4+ after reform	-0.029 (0.047)	0.007 (0.050)	-0.039 (0.054)	0.003 (0.062)	-0.022 (0.043)	0.010 (0.046)	-0.038 (0.048)	-0.006 (0.044)
Response to interstate branch banking deregulations								
Years 1–2 before reform	0.028 (0.032)	-0.016 (0.027)	0.000 (0.037)	-0.031 (0.036)	0.057 (0.032)	-0.005 (0.024)	-0.043 (0.022)	-0.043 (0.019)
Reform year and one after	0.062 (0.034)	0.004 (0.030)	0.028 (0.048)	-0.001 (0.046)	0.096 (0.030)	0.010 (0.027)	-0.020 (0.037)	-0.049 (0.027)
Years 2–3 after reform	0.175 (0.037)	0.078 (0.034)	0.167 (0.052)	0.106 (0.052)	0.189 (0.042)	0.068 (0.029)	0.047 (0.045)	-0.036 (0.042)
Years 4+ after reform	0.227 (0.057)	0.124 (0.046)	0.272 (0.065)	0.152 (0.060)	0.194 (0.060)	0.115 (0.049)	0.163 (0.050)	0.015 (0.046)
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,097	1,097	1,097	1,097	1,097	1,097	1,097	1,097

Table 4: Dynamic state-year panel estimates. The key rows are the interstate post-reform periods, where coefficients grow after reform for start-up entry and closures.

6.6 Table 5: Relative entry count differences-in-differences; Table 6: Dynamic relative entry design

Read: Table 5 compares start-ups with facility expansions in state-industry-type-year cells. Table 6 adds dynamic leads/lags. Interstate deregulation increases start-up entry relative to facility expansions, especially churning entry.

Economic message: These are the cleanest entrepreneurship-specific estimates because they absorb state-industry-year shocks common to start-ups and facility expansions.

Table 5

Differences-in-differences estimates of establishment entry and exit using data from the Longitudinal Business Database (LBD) for 1977–1998. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on the entry and exit of new firms relative to state-industry-type-year establishment expansions and contractions for existing multi-unit firms. Dependent variables are indicated in column headers. Cells are constructed by state-industry-type-year, where type indicates whether the entrant is a new single-unit firm or a new establishment of a multi-unit firm. All regressions include cross-sectional fixed effects for state-industry-type and longitudinal fixed effects for type-year and state-industry-year. In these saturated models, single-unit responses are estimated relative to multi-unit responses. Regressions include unreported indicator variables for cells with zero births and unreported interactions of explanatory variables with census-year indicators. Regressions are weighted by average birth employment in cells from 1977 to 1985. Standard errors are clustered at the cross-sectional state-type level. Pre-post specifications compare annual entry rates before and after the state-level banking deregulation indicated. Linear treatment effect specifications allow for linear growth in treatment effects over time by modeling the number of years after the indicated deregulation's passage, with a long-term effect at four years.

	Log entry counts (1)	Log churning entry (2)	Log long-term entry (3)	Log closure counts (4)
Panel A. Pre-post specifications				
Start-up response relative to multi-unit facilities				
Intrastate banking deregulation post indicator × start-up firms	−0.012 (0.032)	−0.038 (0.038)	0.006 (0.030)	−0.020 (0.028)
Interstate banking deregulation post indicator × start-up firms	0.027 (0.019)	0.041 (0.023)	0.029 (0.017)	0.046 (0.014)
State-industry-type fixed effects	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894
B. Linear treatment effect specifications				
Start-up response relative to multi-unit facilities				
Number of years since intrastate banking reform × start-up firms	−0.010 (0.006)	−0.015 (0.008)	−0.006 (0.007)	−0.007 (0.005)
Number of years since interstate banking reform × start-up firms	0.021 (0.007)	0.038 (0.009)	0.013 (0.007)	0.036 (0.009)
State-industry-type fixed effects	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894

Table 5: relative entry count regressions.

Relative start-up design. Start-ups respond more than facility expansions after interstate reform, isolating entrepreneurial entry from general establishment growth.

6.7 Tables 7 and 8: Size distribution of churning and long-term entry

Read: Table 7 and Table 8 disaggregate start-up responses by entrant employment size. The largest churning and long-term effects are among small entrants, especially firms with 1-5 or 6-20 employees.

Table 6

Dynamic differences-in-differences estimates of establishment entry and exit using data from the Longitudinal Business Database (LBD) for 1977–1998. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on the entry and exit of new firms relative to state-industry-type-year establishment expansions and contractions for existing multi-unit firms. Dynamic specifications model a series of leads and lags for each reform. Leads and lags are consolidated into two-year increments extending from two years prior to the deregulations to four or more years after the deregulations. Coefficient values for dynamic leads and lags are relative to the period three years before reforms and earlier. Table 5 provides further details on the specification.

	Log entry counts (1)	Log churning entry (2)	Log long-term entry (3)	Log closure counts (4)
Start-up response relative to multi-unit facilities following intrastate branch banking deregulations				
Years 1–2 before reform	−0.013 (0.023)	−0.020 (0.035)	−0.016 (0.019)	−0.028 (0.032)
Reform year and one after	−0.020 (0.036)	−0.039 (0.045)	−0.009 (0.028)	−0.049 (0.037)
Years 2–3 after reform	−0.084 (0.036)	−0.135 (0.048)	−0.059 (0.032)	−0.079 (0.041)
Years 4+ after reform	−0.044 (0.030)	−0.061 (0.037)	−0.035 (0.030)	−0.028 (0.023)
Start-up response relative to multi-unit facilities following interstate branch banking deregulations				
Years 1–2 before reform	0.021 (0.034)	0.040 (0.037)	0.025 (0.033)	0.006 (0.022)
Reform year and one after	0.038 (0.034)	0.061 (0.040)	0.044 (0.030)	0.045 (0.028)
Years 2–3 after reform	0.071 (0.039)	0.115 (0.054)	0.067 (0.036)	0.097 (0.040)
Years 4+ after reform	0.109 (0.037)	0.220 (0.054)	0.059 (0.035)	0.173 (0.045)
State-industry-type fixed effects	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894

Table 6: dynamic relative entry regressions.

Economic message: The reforms primarily relax extensive-margin constraints for small entrepreneurs, rather than only financing large new entrants.

Table 7
Table disaggregates churning and long-term entry by employment size of entrant in the first year of operation. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on the entry and exit of new firms relative to state-industry-type year establishment expansions and contractions for existing multi-unit firms. Table 5 provides further details on the specification.

	Log churning entrants by initial employment				Log long-term entrants by initial employment			
	1-5 (1)	6-20 (2)	21-100 (3)	101+ (4)	1-5 (5)	6-20 (6)	21-100 (7)	101+ (8)
Panel A. Pre-post specifications								
Intrastate banking deregulation post indicator × start-up firms	0.005 (0.026)	-0.044 (0.032)	0.015 (0.035)	-0.034 (0.042)	0.048 (0.018)	-0.013 (0.024)	-0.009 (0.032)	-0.034 (0.031)
Interstate banking deregulation post indicator × start-up firms	0.047 (0.019)	0.040 (0.023)	-0.026 (0.027)	0.032 (0.030)	0.028 (0.023)	0.046 (0.022)	-0.003 (0.026)	0.005 (0.022)
State-industry-type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894	111,894	111,894	111,894	111,894
B. Linear treatment effect specifications								
Number of years since intrastate banking reform × start-up firms	-0.002 (0.008)	-0.014 (0.007)	-0.006 (0.007)	-0.013 (0.010)	0.002 (0.007)	-0.008 (0.007)	-0.005 (0.009)	-0.008 (0.008)
Number of years since interstate banking reform × start-up firms	0.038 (0.010)	0.053 (0.012)	0.027 (0.010)	0.012 (0.010)	0.023 (0.010)	0.029 (0.009)	0.022 (0.012)	0.003 (0.013)
State-industry-type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894	111,894	111,894	111,894	111,894

Table 7: static size distribution of entry effects.

Entrant-size effects. The entry boom is concentrated in smaller start-ups, supporting the financial-constraint interpretation.

6.8 Tables 9 and 10: Incumbent market shares and concentration

Read: Tables 9 and 10 examine whether deregulation reduces incumbent market shares or concentration. The estimated effects are small and often statistically weak.

Economic message: The strongest evidence is for democratized entry and churning, not for a large measured decline in incumbent concentration.

Table 9
Panel estimations of incumbent market shares and concentration at state-industry-year level using data from the Longitudinal Business Database (LBD) for 1977-1998. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on industrial concentrations. Dependent variables are indicated in column headers. Cells are constructed by state-industry-year. Regressions include cross-sectional fixed effects for state-industry and longitudinal fixed effects for industry-year. Regressions include unreported interactions of explanatory variables with census-year indicators. Regressions are weighted by average birth employment in cells from 1977 to 1985. Standard errors are clustered at the cross-sectional state-industry level. Incumbent firms are defined as the 10 largest firms in 1982 by state-industry. Market shares are calculated through employments.

	Log top ten incumbent market share (1)	Log top ten incumbent market share (2)	Log top ten incumbent market share (3)	Log top ten incumbent market share (4)	Normalized Herfindahl-Hirschman index (5)	Normalized Herfindahl-Hirschman index (6)
Panel A. Pre-post specifications						
Intrastate banking deregulation post indicator	-0.019 (0.026)	-0.009 (0.012)	0.001 (0.009)	-0.001 (0.005)	-0.001 (0.002)	-0.001 (0.001)
Interstate banking deregulation post indicator	-0.047 (0.033)	-0.044 (0.022)	-0.005 (0.007)	-0.003 (0.006)	0.002 (0.001)	0.001 (0.001)
State-industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	55,947	55,947	55,947	55,947	55,947	55,947
B. Linear treatment effect specifications						
Number of years since intrastate banking reform	0.008 (0.009)	-0.003 (0.010)	0.000 (0.004)	-0.001 (0.004)	0.001 (0.001)	-0.001 (0.001)
Number of years since interstate banking reform	-0.008 (0.010)	-0.009 (0.005)	-0.007 (0.002)	-0.007 (0.002)	-0.001 (0.001)	-0.001 (0.001)
State-industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	55,947	55,947	55,947	55,947	55,947	55,947

Table 9: concentration regressions.

Creative-destruction margin. The paper finds weaker direct evidence of incumbent displacement than of entry and churning.

Table 8
Table disaggregates churning and long-term entry by employment size of entrant in the first year of operation. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on the entry and exit of new firms relative to state-industry-type year establishment expansions and contractions for existing multi-unit firms. Table 6 provides further details on the specification.

	Log churning entrants by initial employment				Log long-term entrants by initial employment			
	1-5 (1)	6-20 (2)	21-100 (3)	101+ (4)	1-5 (5)	6-20 (6)	21-100 (7)	101+ (8)
Start-up response relative to multi-unit facilities following intrastate branch banking deregulations								
Years 1-2 before reform	-0.018 (0.034)	-0.016 (0.040)	-0.040 (0.030)	0.001 (0.037)	-0.023 (0.024)	-0.012 (0.036)	-0.061 (0.032)	-0.073 (0.039)
Reform year and one after	-0.028 (0.029)	-0.041 (0.046)	-0.015 (0.058)	-0.014 (0.068)	0.011 (0.022)	-0.045 (0.033)	-0.084 (0.033)	-0.070 (0.035)
Years 2-3 after reform	-0.081 (0.030)	-0.104 (0.043)	-0.051 (0.040)	-0.140 (0.053)	-0.051 (0.032)	-0.045 (0.033)	-0.038 (0.038)	-0.058 (0.032)
Years 4+ after reform	-0.009 (0.035)	-0.068 (0.031)	0.045 (0.033)	-0.024 (0.049)	-0.004 (0.027)	-0.048 (0.035)	-0.053 (0.038)	-0.054 (0.040)
Start-up response relative to multi-unit facilities following interstate branch banking deregulations								
Years 1-2 before reform	0.053 (0.022)	0.073 (0.031)	0.031 (0.046)	0.012 (0.046)	0.033 (0.031)	0.043 (0.035)	0.063 (0.038)	0.054 (0.031)
Reform year and one after	0.073 (0.022)	0.066 (0.031)	-0.022 (0.042)	0.032 (0.053)	0.036 (0.032)	0.072 (0.026)	0.031 (0.028)	0.040 (0.028)
Years 2-3 after reform	0.135 (0.036)	0.165 (0.045)	0.066 (0.058)	0.049 (0.084)	0.105 (0.042)	0.124 (0.033)	0.092 (0.048)	0.045 (0.037)
Years 4+ after reform	0.226 (0.048)	0.289 (0.058)	0.107 (0.061)	-0.039 (0.063)	0.074 (0.051)	0.174 (0.041)	0.086 (0.056)	0.038 (0.069)
State-industry-type fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Type-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	111,894	111,894	111,894	111,894	111,894	111,894	111,894	111,894

Table 8: dynamic size distribution of entry effects.

Table 10
Dynamic panel estimations of incumbent market shares and concentration at state-industry-year level using data from the Longitudinal Business Database (LBD) for 1977-1998. Coefficients report estimates of the effect of intrastate and interstate banking deregulations on industrial concentrations. Dynamic specifications model a series of leads and lags for each reform. Leads and lags are consolidated into two-year increments extending from two years prior to the deregulations to four or more years after the deregulations. Coefficient values for dynamic leads and lags are relative to the period three years before reforms and earlier. Table 9 provides further details on the specification.

	Log top ten incumbent market share (1)	Log top ten incumbent market share (2)	Log top ten incumbent market share (3)	Log top ten incumbent market share (4)	Normalized Herfindahl-Hirschman index (5)	Normalized Herfindahl-Hirschman index (6)
Response to intrastate branch banking deregulations						
Years 1-2 before reform	0.027 (0.019)	-0.003 (0.013)	-0.004 (0.008)	0.005 (0.008)	0.003 (0.001)	0.001 (0.001)
Reform year and one after	0.023 (0.025)	0.001 (0.018)	0.009 (0.012)	0.016 (0.010)	0.001 (0.002)	-0.001 (0.001)
Years 2-3 after reform	0.005 (0.044)	-0.016 (0.030)	-0.009 (0.016)	0.002 (0.016)	0.001 (0.002)	-0.002 (0.002)
Years 4+ after reform	0.044 (0.037)	-0.008 (0.038)	-0.002 (0.017)	0.000 (0.017)	0.002 (0.003)	-0.001 (0.001)
Response to interstate branch banking deregulations						
Years 1-2 before reform	-0.045 (0.031)	-0.035 (0.016)	-0.008 (0.010)	-0.009 (0.008)	0.000 (0.001)	0.000 (0.001)
Reform year and one after	-0.085 (0.055)	-0.069 (0.029)	-0.006 (0.011)	-0.005 (0.009)	0.002 (0.002)	0.002 (0.001)
Years 2-3 after reform	-0.092 (0.065)	-0.078 (0.032)	-0.027 (0.013)	-0.028 (0.010)	0.000 (0.002)	0.000 (0.001)
Years 4+ after reform	-0.074 (0.077)	-0.073 (0.041)	-0.025 (0.016)	-0.026 (0.012)	0.000 (0.002)	0.001 (0.002)
State-industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Table 10: dynamic concentration regressions.

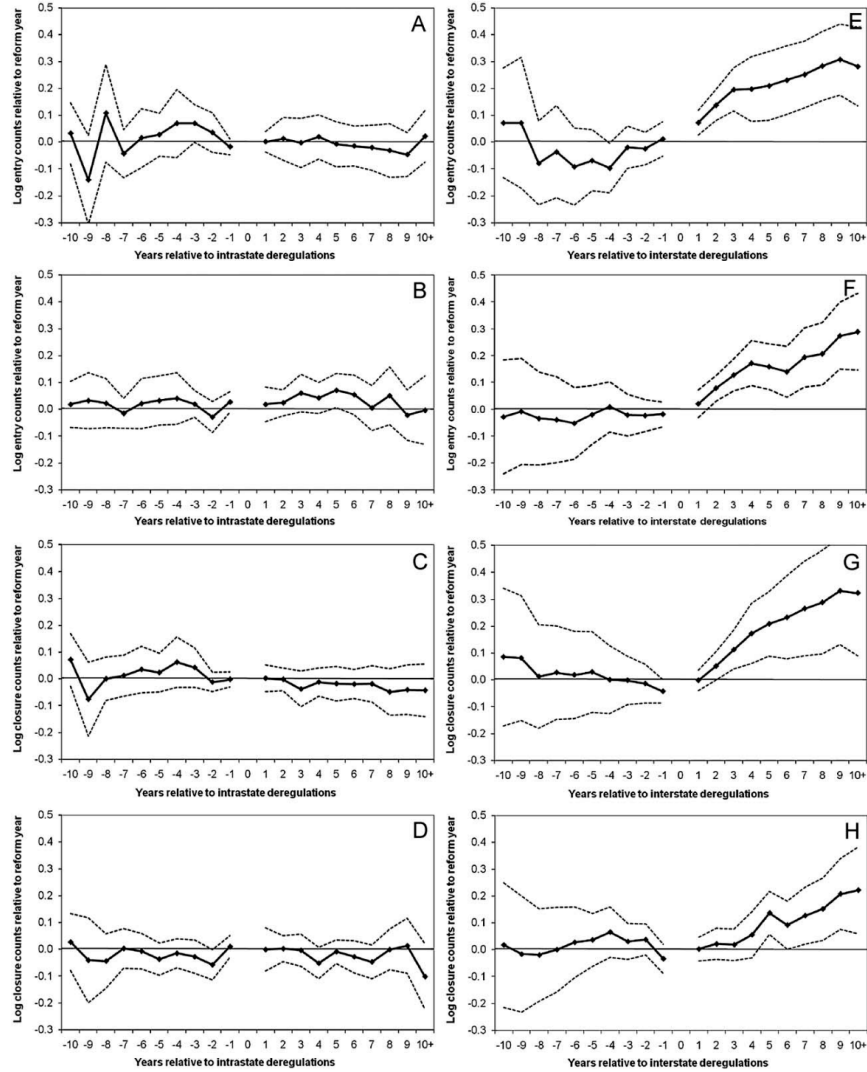


Fig. A1. Single-unit and multi-unit dynamics for entry and closures. Panel A. Intrastate, SU birth, Panel B. Intrastate, MU birth, Panel C. Intrastate, SU

Fig. A1: Separate single-unit and multi-unit dynamics for entry and closures. This appendix figure supports the distinction between entrepreneurship and facility expansion.

6.9 Fig. A1: Single-unit and multi-unit dynamics

Read: Appendix Fig. A1 separates single-unit and multi-unit establishment dynamics. The single-unit panels are most relevant to entrepreneurship, while multi-unit panels reflect expansion or contraction by existing firms.

Economic message: The appendix visual reinforces the paper's interpretation: the treatment effects are strongest for new single-unit start-ups and their closures.

7 Short summary

- U.S. branch banking deregulation relaxed financial constraints and changed local credit markets.
- Interstate deregulation has much stronger effects than intrastate deregulation.

- Entry rises after interstate reform, especially among new single-unit start-ups.
- Much of the increase is churning entry: firms that close within three years.
- Long-term entry rises more modestly, implying some genuine durable entrepreneurship.
- Closures also increase, but direct evidence of incumbent displacement is weaker.
- Effects are concentrated among smaller entrants, consistent with relaxing financing constraints for small entrepreneurs.
- The paper’s central message is democratized experimentation rather than perfect ex ante capital allocation.

8 Conclusion

8.1 Core conclusion

Kerr and Nanda show that banking deregulation democratized entrepreneurial entry. Better banking markets allowed more people to start firms, but the resulting entry boom included many failures. The process of creative destruction therefore operated through experimentation, not through banks perfectly selecting only the best projects ex ante.

8.2 Economic intuition

When credit access expands, marginal entrepreneurs can enter. Some are productive and survive; many are not and quickly exit. This churn is not a side issue - it is part of the entrepreneurial discovery process. Financial development raises the option value of trying.

8.3 Incremental contribution revisited

The paper’s key contribution is to combine banking deregulation variation with establishment-level data and survival categories. This lets the authors show that the finance-entrepreneurship channel works through both successful entrants and failed attempts. Without distinguishing churning from long-term entry, one would miss the central mechanism.

8.4 Implications

The paper cautions against evaluating financial development only by success rates or aggregate growth. A financial system that supports entrepreneurship may look inefficient in the short run because it produces many failures. But those failures are the cost of broad access and discovery.

8.5 Limitations

The paper does not observe individual loan decisions or entrepreneur credit constraints directly. The results are reduced-form effects of banking deregulation. Deregulation may also have affected demand, competition, and bank consolidation. The paper addresses these concerns with fixed effects, event dynamics, relative comparisons, and external-finance interactions, but the mechanism remains indirectly inferred.

8.6 Takeaway

The enduring lesson is that democratizing finance democratizes experimentation. Financial reforms can expand entrepreneurship not by eliminating failure, but by allowing more potential entrepreneurs to test ideas in the market.